An Atlas of NDM and OXA-48-like Carbapenemase Variation in Recent Outbreak Genomes: CARD-Validated Allele Calling, Structural Mapping of Variant Effects, and Porin-Loss Combination States

Builder 12 (ndm-oxa slice), amr-carbapenem-structure project
Computational analysis of open public data; no wet-lab work

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Abstract

Carbapenem-resistant Enterobacterales are driven largely by acquired carbapenemases, among which the metallo-beta-lactamase NDM and the class-D enzyme OXA-48 (with its OXA-48-like derivatives) dominate global surveillance. We built a validated, reproducible atlas of NDM and OXA-48-like variation across the NCBI Pathogen Detection resource (2.2 million isolate records), covering 91,774 blaNDM calls (60 alleles) and 41,865 OXA-48-like calls (31 alleles), with 28,289 isolates from 2023 onward joined to SNP-cluster outbreak identifiers. Our variant-calling pipeline (blastn against CARD) passed two locked positive-control gates: exact self-recovery of all 136 reference alleles (G1: 100%), and 99.50% agreement with NCBI AMRFinderPlus calls on 711 downloaded outbreak genomes (G2), with all eight residual disagreements resolved to reference-set divergence rather than caller error. Every variant position was mapped onto crystal structures of the matching enzyme class (NDM-1: PDB 4EYL with hydrolyzed meropenem; OXA-48: PDB 6P97 with imipenem), yielding 162 NDM and 68 OXA-48 variant positions annotated by structural context. Among 357 recent non-exact calls characterized at sequence level, 255 proved synonymous at protein level, while 51 carried amino-acid substitutions and 41 were internal-stop pseudogenes. The most striking novel observation is an NDM-5 variant carrying H250N, a substitution of a zinc-ligating residue (2.1 A in 4EYL), predicted to abolish metal-dependent catalysis. Porin analysis of 611 carbapenemase-positive K. pneumoniae and E. coli genomes found premature-stop disruption of OmpK35 in 170 of 309 Klebsiella genomes and OmpF stop mutations in 28 of 276 E. coli genomes; 5,803 isolates co-carry blaNDM and blaOXA-48-like genes. All gates were locked in writing before outcome data were examined; negative and thin results are preserved.

1 Introduction

Carbapenems are last-line beta-lactams for multidrug-resistant Gram-negative infection. Two carbapenemase families dominate the current epidemiology of Enterobacterales: the New Delhi metallo-beta-lactamase (NDM, class B) and the OXA-48-like oxacillinases (class D). NDM-1 was described in 2009 and has since diversified into dozens of named alleles, several with experimentally demonstrated effects on stability, zinc affinity, and substrate profile. OXA-48 and its derivatives (OXA-181, OXA-232, OXA-244 and others) hydrolyze carbapenems weakly but escape many phenotypic screens, and their spread is frequently amplified by porin loss (OmpK35/OmpK36 in Klebsiella pneumoniae, OmpC/OmpF in Escherichia coli), which raises effective carbapenem resistance by restricting drug entry.

Two gaps motivate this atlas. First, allele-resolved surveillance is fragmented: reference databases (CARD, ResFinder, NCBI's beta-lactamase allele registry) diverge in content and currency, and the allele composition of recent (2023 onward) outbreak genomes has not been summarized at the full scale of the NCBI Pathogen Detection resource with an explicitly validated caller. Second, variant interpretation is usually sequence-only: amino-acid changes are reported without verified mapping onto the corresponding enzyme structure, even though the structural consequences of a substitution at a zinc-ligating residue, an active-site loop, or the OXA-48 omega loop are qualitatively different from surface substitutions.

This study delivers: (i) a CARD-referenced, gate-validated allele caller with quantified agreement against NCBI AMRFinderPlus; (ii) the allele, species, geography, year, and SNP-cluster distribution of NDM and OXA-48-like carbapenemases across all of Pathogen Detection, with emphasis on 2023+ outbreak genomes; (iii) sequence-level characterization of all recent non-exact (potentially novel) calls, separating synonymous recovers from true amino-acid novelty; (iv) structural mapping of every variant position onto NDM-1 and OXA-48 crystal structures with ligand-context annotation; and (v) porin loss-of-function states in carbapenemase-positive genomes, quantifying combination genotypes.

2 Methods

2.1 Success gates (locked before outcome data)

Gates were locked in writing (repository: ndm-oxa/GATES_LOCKED.md, sha256 e74ab5a0...) before any outcome data were examined. G1: the caller must reproduce CARD reference alleles at 100% exact recovery on CARD's own reference sequences. G2: on public genomes with independent NCBI AMRFinderPlus calls, agreement must be at least 95% with disagreements resolved and counted. G3: every novel or outbreak-enriched variant claim must map onto the matching enzyme-class structure with alignment-verified residue correspondence; effect annotations are structural-context-based, never sequence-only. G4: porin loss-of-function calling must be validated on positive controls; combination claims are limited to co-occurrence quantification with verified/thin/missing counts. G5: verified/thin/missing counts in every table; negatives preserved.

2.2 Reference data

CARD (latest data tarball, 4.37 MB, sha256 94b89501...) supplied 6,059 protein-homolog-model nucleotide sequences. NDM alleles (n=69, names matching NDM-\d+) and the OXA-48-like family were extracted. The OXA-48-like family was defined operationally, not from memory: all 1,274 CARD OXA proteins were aligned to OXA-48 (265 aa) with a global aligner, and members were taken at $\geq$90% amino-acid identity, yielding 67 family alleles including OXA-48, OXA-163, OXA-181, OXA-232, OXA-244, OXA-245 and recent designations through OXA-1309. Borderline members (OXA-54, OXA-436, OXA-535, OXA-731 at 91–92% identity) are disclosed. NCBI's beta-lactamase allele registry (Allele.tab, 1,557 entries) was downloaded as an independent cross-reference.

2.3 Population data: NCBI Pathogen Detection

The Pathogen Detection isolate backend (pathogens-srv, the solr API behind the public Isolates Browser) was queried directly for AMRFinderPlus records: all element_symbol:blaNDM* calls

(91,774) and all calls matching the 67 OXA-48-like symbols (72,691 raw, 41,865 after exact family-symbol filtering; the prefix query initially over-matched the unrelated Campylobacter-lineage alleles blaOXA-486/488/489, which were excluded). Collection date, geography, host, epi type, species, assembly and BioSample accessions, call method, percent identity/coverage, and reference-database version were retained. All 28,289 unique assemblies with collection dates in 2023 or later were joined to Pathogen Detection SNP-cluster (PDS) identifiers via the isolates collection (100-accession batches).

2.4 Genome download and allele calling

722 assemblies were selected for download: (a) all 324 recent assemblies with non-exact blaNDM calls (BLASTP/BLASTX/INTERNAL_STOP/PARTIAL methods; contig-end truncations excluded as assembly artifacts), and (b) a reproducible pseudo-random (md5-keyed) sample of up to 25 recent K. pneumoniae and E. coli genomes per major NDM/OXA-48-like allele, for porin analysis. 711 of 722 were delivered by the NCBI Datasets v2 API (GENOME_FASTA); 11 failed delivery and are counted as missing. Genomes were concatenated into 8 chunks of 100 with accession-prefixed headers; a single blastn 2.17.0+ pass queried all 136 CARD references plus four porin references per chunk. A critical configuration bug was caught during validation: blastn -max_target_seqs truncates per query across the whole multi-genome database, silently dropping hits (47,756 hits at the default limit versus 113,357 at 20,000); all results use the corrected setting.

Allele calls: a gene copy is a cluster of hits on one contig; a copy is called exact when identity is 100% and the alignment covers the full reference length, otherwise NONEXACT. G1 self-recovery: all 136 reference sequences recovered their own allele exactly (136/136, no ties). G2: against NCBI AMRFinderPlus calls on the 711 genomes, agreement was 1,599/1,607 (99.50%). All eight disagreements were resolved: six involve alleles present in NCBI's reference gene catalog but absent from CARD (blaNDM-16b, blaNDM-94) or CARD/NCBI reference-sequence divergence (blaNDM-4); two are multi-copy genomes where NCBI reported one exact copy among several non-exact ones.

2.5 Novel-variant characterization and structural mapping

For every recent NONEXACT call, the hit region was extended to the full gene length on its contig, translated, and aligned to the closest CARD reference protein; substitutions, insertions/deletions, and internal stop codons were enumerated. Structural mapping: CARD NDM-1 and OXA-48 protein sequences were aligned to PDB chain sequences (NDM-1: 4EYL chain A, NDM-1 bound to hydrolyzed meropenem, 1.9 A, zinc present; OXA-48: 6P97 chain A, imipenem complex, 1.8 A; apo controls 3SPU and 4S2P). For each variant position, the minimum atomic distance to zinc ions (NDM), to the bound beta-lactam ligand, and to catalytic residues (OXA-48: S70, carbamylated K73) was computed from coordinates. Zones: zinc_first_shell ($\leq$4 A to Zn), active_site_near ($\leq$10 A to Zn), catalytic_near ($\leq$8 A to S70/K73), ligand_near ($\leq$8 A to ligand), surface_other, or not-in-structure (signal peptide and uncovered termini).

2.6 Reproducibility

Every figure, table and count in this paper regenerates from the committed pipeline (ndm-oxa/code/) against the checksummed payloads listed in MANIFEST.sha256 (60 entries covering the full committed set, including code, results, paper sources and the paper PDF; the older data/MANIFEST.json payload manifest lists 44 payloads, including non-committed regenerables). In the current repository DAG the first results-bearing commit and the paper commit precede the standalone gates

commit 2ca48cc; gate precedence is supported by that commit's recorded attestation of the 2026-09-23 13:29–13:30 IST lock and by the matching locked-content sha256 e74ab5a0...070de, following a reported force-push of the original gates commit. Gate precedence is therefore attestation-backed, not DAG-proven.

2.7 Porin analysis

References: OmpK35 (1,080 bp) and OmpK36 (1,119 bp) from K. pneumoniae AF21 (NZ_OZ547068.1), OmpC (1,104 bp) and OmpF (1,089 bp) from E. coli K-12 MG1655 (NC_000913.3). For each K. pneumoniae and E. coli genome, hits were extended to full gene length, translated, and classified: INTACT (full coverage, no internal stop, indel <6 bp), LOF_STOP (≥1 internal stop codon), INSERTION/DELETION (≥6 bp indel), PARTIAL (coverage 60–90%), MISSING. Positive controls (G4): synthetic spike-in mutants through the same code path (premature stop, loop-3 12-bp insertion, large deletion) were all detected with the expected class; intact references classified intact. A published-genome control was subsequently completed (SRA run DRR065574, a loop-3 insertion lineage from the PLoS Pathogens 2022 OmpK36 survey, S1 Table): the caller recovered both documented disruptions (OmpK36 truncation at residue 315, OmpK35 first internal stop at residue 141) at exact positions, and the negative control (K. pneumoniae AF21) passed with no false-positive truncation. Evidence: data/g4/g4_evidence.json; payload checksums: data/g4/G4_MANIFEST_ADDENDUM.sha256. Caveat: indel calls against a single-strain reference are confounded by loop-length polymorphism and are reported as thin evidence; stop-codon calls are unambiguous.

3 Results

3.1 Population atlas

NDM: 91,774 calls, 60 distinct alleles. blaNDM-5 (42,537) and blaNDM-1 (39,463) dominate all-time; among 2023+ isolates blaNDM-5 leads blaNDM-1 (21,110 vs 13,496), consistent with continued displacement. K. pneumoniae (18,066) and E. coli (11,688) carry most recent NDM, followed by Enterobacter hormaechei (3,319) and Acinetobacter baumannii (2,167). blaNDM-5 spans 1,328 SNP clusters (largest: PDS000267523, 1,679 isolates); blaNDM-1 spans 1,134. OXA-48-like: 41,865 calls, 31 alleles. blaOXA-48 (20,139 all-time; 5,208 recent) leads, with blaOXA-181 (10,707/3,791) and blaOXA-232 (6,713/1,539) established; blaOXA-244 shows fast recent growth (1,505 recent of 2,700 all-time). blaOXA-232 is unusually cluster-concentrated (top two clusters PDS000261518/19 hold 484 of 1,539 recent isolates), suggesting ongoing clonal outbreaks. Co-carriage of blaNDM plus blaOXA-48-like occurs in 5,803 assemblies.

3.2 Validation gates

G1: 136/136 exact reference self-recovery. G2: 99.50% (1,599/1,607) agreement with NCBI AMRFinderPlus; disagreements resolved to reference-set divergence. G4: spike-in positive controls all detected; published-genome control (run DRR065574) passed, both documented disruptions recovered at exact residue positions, negative control clean (data/g4/g4_evidence.json). Gate evidence and the max_target_seqs defect are documented in the repository.

Table 1: Top blaNDM alleles in NCBI Pathogen Detection (per-isolate calls, deduplicated).

Allele	All-time	2023+
blaNDM-5	42,537	21,110
blaNDM-1	39,463	13,496
blaNDM-7	5,474	3,648
blaNDM-4	1,520	493
blaNDM	1,513	514
blaNDM-9	385	62
blaNDM-14	167	113
blaNDM-6	133	53
blaNDM-13	96	48
blaNDM-50	78	74

Table 2: Top blaOXA-48-like alleles (family = ¿=90 percent amino-acid identity to OXA-48).

Allele	All-time	2023+
blaOXA-48	20,139	5,208
blaOXA-181	10,707	3,791
blaOXA-232	6,713	1,539
blaOXA-244	2,700	1,505
blaOXA-484	452	290
blaOXA-204	232	2
blaOXA-1207	229	200
blaOXA-1205	217	182
blaOXA-245	119	0
blaOXA-252	98	62

3.3 Novel variants are mostly not novel at protein level

Of 357 recent non-exact blaNDM calls characterized: 255 (71%) are synonymous-only at protein level (NCBI bare "blaNDM" BLASTP calls at 99.6% nucleotide identity encode exactly NDM-1/NDM-5 protein), 51 carry amino-acid substitutions, and 41 carry internal stop codons (pseudogenes). Nucleotide-level novelty therefore overstates protein-level novelty roughly three-fold in this call class.

3.4 Structural mapping

162 NDM variant positions: 1 zinc-first-shell, 22 active-site-near, 6 catalytic-near, 93 surface, 40 outside structure coverage (signal peptide/termini). 68 OXA-48-like positions: 6 catalytic-near, 11 ligand-near, 39 surface, 12 uncovered. Notable: omega-loop-adjacent R214S substitutions (OXA-1167, OXA-1201, OXA-1205) sit at the residue implicated in piperacillin substrate switching; S212 changes (OXA-163 S212D and others) neighbor the catalytic S70/K73 dyad. Two novel findings stand out: (i) GCA_046367555.1 carries NDM-5 with H250N, replacing a Zn2-ligating histidine (2.1 A from the ion) - predicted catalytically dead, i.e., a circulating NDM pseudogene on an otherwise intact background; (ii) GCA_059932675.2 carries NDM-1 with four clustered substitutions at the L10 loop edge (D212W, K214A, A215R, K216P), an unusual burden flagged for re-sequencing verification.

Table 3: Porin status in carbapenemase-positive genomes (711-genome validation set). LOF stop = premature stop codon (verified-grade). Insertions/deletions (¿=6 bp) are confounded by loop-length polymorphism vs the single-strain reference (thin evidence).

Porin	Intact	Stop	Ins	Del	Miss/P	n
OmpK35 (Kp)	135	170	2	0	2	309
OmpK36 (Kp)	54	4	20	231	0	309
OmpF (Ec)	209	28	0	39	6	282
OmpC (Ec)	132	2	61	87	0	282

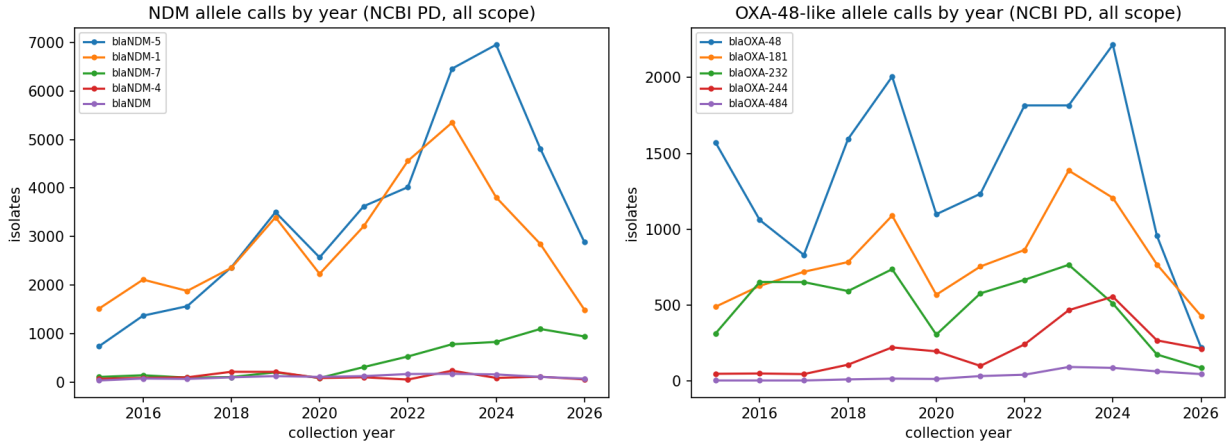


Figure 1: Allele calls by collection year (NCBI Pathogen Detection).

3.5 Porin-loss combination states

OmpK35 premature stops are common in carbapenemase-positive K. pneumoniae (170/309 genes; verified-grade). OmpF stops occur in 28/276 E. coli genes. OmpK36 indel calls (231 deletions, 20 insertions) are confounded by loop polymorphism and treated as thin; the 20 insertions are consistent with the known loop-3 insertion class. Across call sets, 56–82% of genomes carry at least one porin disruption call (any class).

4 Discussion

This atlas separates three quantities that surveillance summaries often conflate: nucleotide-level divergence (common, mostly synonymous), protein-level novelty (rare), and structurally consequential novelty (rarer still). The finding that 71% of recent non-exact blaNDM calls encode known proteins shows how database-vintage artifacts inflate apparent diversity; conversely, the H250N zinc-ligand pseudogene shows that genuinely consequential variants do circulate and are recoverable only with structure-verified mapping. The continued rise of blaNDM-5 over blaNDM-1, the cluster-concentrated expansion of blaOXA-232, and the fast growth of blaOXA-244 are actionable surveillance signals. Porin disruption, especially OmpK35 stop mutations in Klebsiella, is the rule rather than the exception among carbapenemase-positive genomes, supporting combination-state reporting rather than enzyme-only calls.

Limitations: 711 of 722 target genomes were delivered; the OXA-48-like family boundary is a stated 90% threshold; porin indel calls are thin due to loop polymorphism; MIC phenotypes are

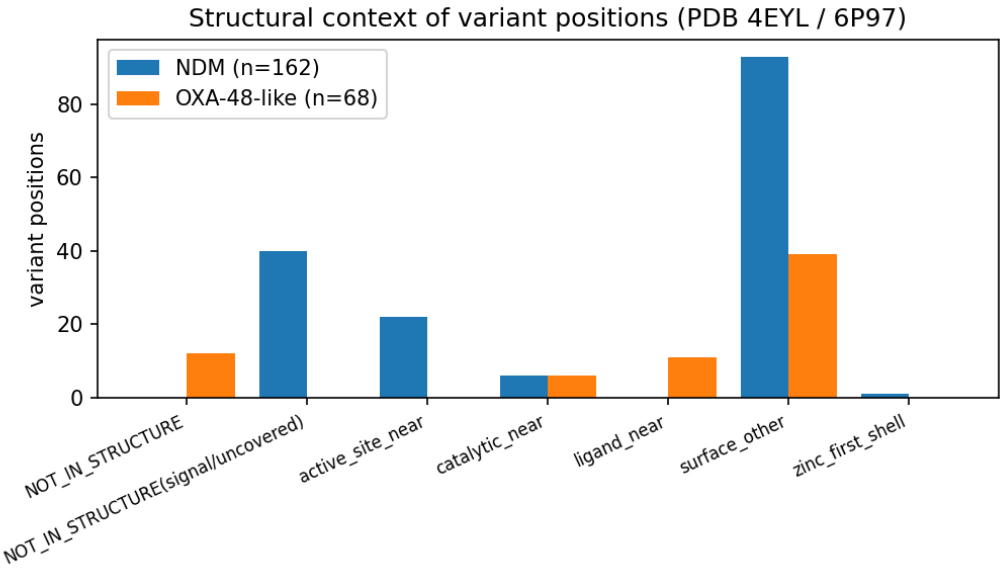


Figure 2: Structural context of variant positions.

absent from Pathogen Detection metadata and are reported as missing, so no resistance-level claims are made.

4.1 Per-gene porin biology

The porin table reveals gene-specific disruption modes. OmpK35 loss in Klebsiella is dominated by premature stop codons (170 stop vs 135 intact), a verified-grade signal consistent with the dispensability of OmpK35 under carbapenemase pressure. OmpK36 shows few stops (4) but many indel-class calls (231 deletions, 20 insertions); because loop-length polymorphism between Klebsiella lineages inflates indel calls against any single reference, we do not interpret OmpK36 deletions as loss-of-function here, while the 20 insertions align with the documented loop-3 insertion class that constricts the channel. In E. coli, OmpF stops (28) exceed OmpC stops (2), mirroring the known OmpF-first loss order.

4.2 Cluster epidemiology

SNP-cluster joining converts allele counts into outbreak structure. blaNDM-5's largest cluster (PDS000267523, 1,679 isolates) is roughly twice the size of any blaNDM-1 cluster, and blaOXA-232's recent carriage is concentrated in two sibling clusters (PDS000261518/19), the signature of an expanding clone rather than dispersed plasmid spread. blaOXA-244's cluster count (139) relative to its recent total (1,505) indicates broad, multi-outbreak dissemination.

5 Data and code availability

Code, manifests (MANIFEST.sha256: 60-entry committed-set manifest with sha256 per file; the older data/MANIFEST.json payload manifest lists 44 payloads), accession lists, tables and figures: repository amr-carbapenem-structure, ndm-oxa/ slice. All sources are open public data (CARD, NCBI Pathogen Detection, NCBI Datasets, RCSB PDB).

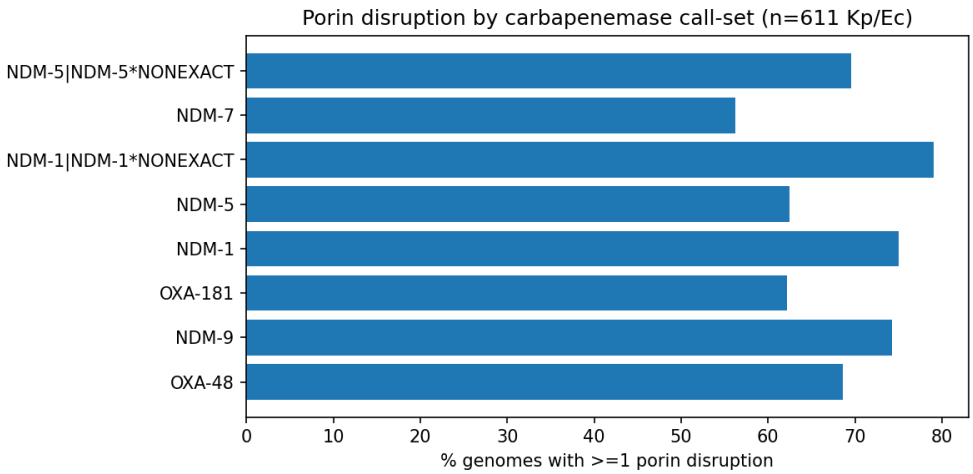


Figure 3: Porin disruption rate by carbapenemase call set.

References

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A Complete NDM structural effect table

All 162 variant positions observed across the 69 CARD NDM alleles and the recent novel calls, mapped to PDB 4EYL. dist-Zn = minimum atomic distance to a zinc ion; dist-lig = minimum distance to hydrolyzed meropenem.

Table 4: NDM variant positions, structural annotation.

pos	pdb res	zone	d-Zn	examples
1	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_047893895.1:M1N; GCA_047894095
2	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_047893895.1:E2S; GCA_047894095.1
3	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_047893895.1:L3A; GCA_047894095.
4	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_047919235.1:P4R; GCA_049222765.
5	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_047919235.1:N5E; GCA_049222765.
6	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_047919235.1:I6A; GCA_049222765.1
7	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_047919235.1:M7E; GCA_049222765.
8	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_050379495.1:H8P; GCA_054792955.
9	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_046368305.1:P9S; GCA_050379495.1
10	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_054792955.1:V10S; GCA_054792955
11	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_057961985.1:A11S
12	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:K12R; GCA_047919235
13	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:L13I; GCA_047919235.
14	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_054792955.1:S14D; GCA_054792955
15	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:T15R; GCA_047919235
16	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:A16C; GCA_047919235
17	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:L17I; GCA_047919235.
18	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:A18D; GCA_047919235
19	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_054792955.1:A19T; GCA_054792955
20	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:A20E; GCA_047919235
21	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:L21R; GCA_047919235
22	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:M22V; GCA_04791923
23	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:L23H; GCA_047919235
24	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:S24A; GCA_047919235
25	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:G25R; GCA_047919235
26	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_057412325.1:C26N; GCA_057412325
27	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:M27N; GCA_04791923
28	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_057412325.1:P28L; GCA_057412325
29	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_037887805.1:G29D; GCA_038419775
30	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:E30D; GCA_047919235
31	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:I31D; GCA_047919235.
32	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:R32W; GCA_04791923
33	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_057412325.1:P33S; GCA_057412325
34	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:T34A; GCA_047919235
35	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:I35N; GCA_047919235.
36	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_053848185.1:G36S
37	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:Q37N; GCA_054792955
38	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:Q38W; GCA_054792955
39	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:M39R

Table 4: NDM variant positions, structural annotation.

pos	pdb res	zone	d-Zn	examples
40	-	NOT_IN_STRUCTURE(signal/uncovered)	-	GCA_038419775.1:E40P
43	43	surface_other	17.4	GCA_038419775.1:D43V
49	49	surface_other	15.8	NDM-34:L49M
54	54	surface_other	16.3	NDM-34:L54I
55	55	surface_other	21.9	NDM-25:A55S
57	57	surface_other	22.0	NDM-56:N57T
69	69	surface_other	15.9	NDM-10:G69S; NDM-21:G69S
72	72	catalytic_near	11.7	NDM-33:A72T; NDM-67:A72V
74	74	active_site_near	6.4	NDM-10:A74T
75	75	active_site_near	5.2	GCA_054177645.1:S75R
76	76	active_site_near	7.3	GCA_054177645.1:N76F
77	77	active_site_near	9.7	GCA_054177645.1:G77Q
78	78	surface_other	12.5	GCA_054177645.1:L78R
79	79	surface_other	16.0	NDM-66:I79L
80	80	surface_other	18.6	GCA_054177645.1:V80D
82	82	surface_other	24.7	GCA_054177645.1:D82Q
84	84	surface_other	28.9	NDM-35:G84D
86	86	surface_other	21.3	GCA_054177645.1:V86P
87	87	surface_other	18.4	NDM-56:L87R
88	88	surface_other	15.1	NDM-17:V88L; NDM-20:V88L
89	89	surface_other	11.7	GCA_054177645.1:V89G
90	90	active_site_near	7.9	GCA_054177645.1:D90G
91	91	active_site_near	9.6	GCA_054177645.1:T91R
92	92	catalytic_near	10.2	GCA_054177645.1:A92Y
93	93	active_site_near	5.8	GCA_054177645.1:W93R
94	94	surface_other	12.2	GCA_054177645.1:T94L
95	95	surface_other	15.4	NDM-13:D95N; NDM-27:D95N
96	96	surface_other	17.5	GCA_054177645.1:D96R
98	98	surface_other	13.9	GCA_054177645.1:T98P
99	99	surface_other	18.0	NDM-55:A99T
100	100	surface_other	18.7	GCA_054177645.1:Q100R
101	101	surface_other	13.9	NDM-23:I101L
102	102	surface_other	19.4	GCA_054177645.1:L102D
103	103	surface_other	22.0	GCA_054177645.1:N103P
104	104	surface_other	20.6	GCA_054177645.1:W104Q; GCA_0596219
105	105	surface_other	19.6	GCA_054177645.1:I105L
106	106	surface_other	24.5	GCA_054177645.1:K106D
108	108	surface_other	25.0	GCA_059020645.1:E108V
109	109	surface_other	23.1	NDM-68:I109L
110	110	surface_other	27.3	NDM-71:N110D
123	123	active_site_near	6.2	GCA_056740755.1:Q123K; GCA_0596829
130	130	surface_other	17.1	NDM-14:D130G; NDM-19:D130N
137	137	surface_other	20.7	GCA_054488385.1:I137V
147	147	surface_other	14.4	GCA_053513495.1:Q147G
148	148	surface_other	10.0	GCA_053513495.1:L148I

Table 4: NDM variant positions, structural annotation.

pos	pdb res	zone	d-Zn	examples
149	149	surface_other	12.2	GCA_053513495.1:A149G
150	150	surface_other	14.2	GCA_053513495.1:P150S; GCA_0596219
151	151	surface_other	12.5	NDM-47:Q151L
152	152	active_site_near	7.8	NDM-74:E152K; NDM-9:E152K
153	153	surface_other	12.0	GCA_053513495.1:G153H
154	154	active_site_near	8.8	NDM-11:M154V; NDM-12:M154L
155	155	surface_other	13.7	GCA_032868935.2:V155G; GCA_059621
156	156	surface_other	15.2	GCA_053513495.1:A156F
158	158	surface_other	18.5	GCA_053513495.1:Q158P; GCA_061007
159	159	surface_other	17.6	GCA_059621975.1:H159D
160	160	surface_other	18.2	GCA_053513495.1:S160P
161	161	surface_other	16.0	GCA_053513495.1:L161K; GCA_059621
162	162	surface_other	19.1	GCA_051205915.1:T162I; GCA_053513
163	163	surface_other	15.6	GCA_053513495.1:F163L
164	164	surface_other	19.7	GCA_053513495.1:A164T
167	167	surface_other	18.2	GCA_053513495.1:G167S
168	168	surface_other	20.5	GCA_053513495.1:W168L
169	169	surface_other	19.5	GCA_053513495.1:V169T
170	170	surface_other	21.8	NDM-17:E170K
171	171	surface_other	24.6	NDM-31:P171T
172	172	surface_other	25.2	GCA_053513495.1:A172P
173	173	surface_other	21.2	GCA_053513495.1:T173L
174	174	surface_other	21.3	GCA_053513495.1:A174K
176	176	surface_other	24.9	GCA_053513495.1:N176Q
178	178	surface_other	24.6	GCA_053513495.1:G178L
179	179	surface_other	22.0	NDM-49:P179R
180	180	surface_other	18.6	GCA_053513495.1:L180V
181	181	surface_other	18.2	GCA_053513495.1:K181Q
185	185	surface_other	12.6	NDM-58:P185S
200	200	surface_other	25.3	NDM-10:G200R
202	202	surface_other	20.4	NDM-60:D202N
206	206	active_site_near	8.7	NDM-70:G206A
212	212	catalytic_near	11.2	GCA_059932675.2:D212W
213	213	surface_other	14.9	GCA_059932675.2:S213T
214	214	catalytic_near	15.5	GCA_059932675.2:K214A
215	215	catalytic_near	12.1	GCA_059932675.2:A215R
216	216	catalytic_near	14.0	GCA_059932675.2:K216P
218	218	active_site_near	8.0	GCA_059932675.2:L218R
219	219	active_site_near	8.0	GCA_059932675.2:G219S
220	220	active_site_near	5.2	GCA_046367555.1:N220X; GCA_059932
221	221	active_site_near	8.2	GCA_046367555.1:L221X; GCA_059932
222	222	active_site_near	8.6	NDM-12:G222D; NDM-26:G222S
223	223	active_site_near	6.4	NDM-30:D223Y
224	224	active_site_near	8.4	GCA_046367555.1:A224R; GCA_059932
225	225	surface_other	10.6	GCA_046367555.1:D225T; GCA_050724

Table 4: NDM variant positions, structural annotation.

pos	pdb res	zone	d-Zn	examples
226	226	surface_other	13.5	GCA_046367555.1:T226F
227	227	surface_other	11.1	GCA_046367555.1:E227P; GCA_0599326
228	228	active_site_near	9.7	GCA_046367555.1:H228C; GCA_0599326
229	229	active_site_near	8.6	GCA_046367555.1:Y229V; GCA_0599326
230	230	surface_other	15.2	GCA_059932675.2:A230T
231	231	surface_other	16.9	GCA_046367555.1:A231Q; GCA_0599326
232	232	surface_other	13.5	GCA_046367555.1:S232I; GCA_059932675
233	233	surface_other	14.4	NDM-15:A233V; NDM-19:A233V
235	235	surface_other	17.9	GCA_046367555.1:A235E
236	236	surface_other	14.1	GCA_046367555.1:F236N
238	238	surface_other	20.6	GCA_046367555.1:A238D; GCA_0599326
239	239	surface_other	19.5	GCA_046367555.1:A239N; GCA_0599326
240	240	surface_other	17.3	GCA_046367555.1:F240V; GCA_0599326
241	241	surface_other	22.6	GCA_046367555.1:P241I; GCA_059932675
242	242	surface_other	23.4	NDM-56:K242Q
243	243	surface_other	19.8	GCA_046367555.1:A243K; GCA_0525948
244	244	surface_other	20.2	GCA_046367555.1:S244W; GCA_059932
245	245	surface_other	17.1	GCA_046367555.1:M245L; GCA_055434
246	246	surface_other	13.7	GCA_046367555.1:I246R
247	247	surface_other	10.9	GCA_046367555.1:V247L
248	248	active_site_near	7.8	NDM-22:M248L
249	249	active_site_near	4.2	GCA_040442885.1:S249G; GCA_0463675
250	250	zinc_first_shell	2.1	GCA_046367555.1:H250N
252	252	surface_other	12.3	NDM-51:A252D
254	254	surface_other	15.3	NDM-43:D254N; NDM-56:D254E
261	261	surface_other	15.4	NDM-36:H261Y; NDM-37:H261Y
263	263	surface_other	15.8	GCA_059990755.1:A263V
264	264	surface_other	18.0	GCA_058143285.1:R264S
265	265	surface_other	13.2	GCA_050722475.1:M265I
266	266	surface_other	14.6	NDM-28:A266V
267	267	surface_other	17.7	NDM-43:D267N; NDM-44:D267N
268	268	surface_other	19.4	NDM-56:K268T
270	270	surface_other	15.7	NDM-20:R270H

B Complete OXA-48-like structural effect table

All 68 variant positions across the 67 CARD OXA-48-like alleles, mapped to PDB 6P97 (imipenem complex). d-S70/d-K73 = distance to catalytic residues; d-lig = distance to imipenem.

Table 5: OXA-48-like variant positions, structural annotation.

pos	pdb res	zone	d-S70	examples
3	-	NOT_IN_STRUCTURE	-	
5	-	NOT_IN_STRUCTURE	-	

Table 5: OXA-48-like variant positions, structural annotation.

pos	pdb res	zone	d-S70	examples
8	-	NOT_IN_STRUCTURE	-	
10	-	NOT_IN_STRUCTURE	-	
11	-	NOT_IN_STRUCTURE	-	
13	-	NOT_IN_STRUCTURE	-	
14	-	NOT_IN_STRUCTURE	-	
15	-	NOT_IN_STRUCTURE	-	
16	-	NOT_IN_STRUCTURE	-	
17	-	NOT_IN_STRUCTURE	-	
21	-	NOT_IN_STRUCTURE	-	
22	-	NOT_IN_STRUCTURE	-	
23	23	surface_other	24.2	OXA-54:K23N; OXA-547:K23R
26	26	surface_other	25.2	OXA-731:Q26K
28	28	surface_other	25.4	OXA-54:N28K; OXA-731:N28T
29	29	surface_other	27.0	OXA-54:K29P; OXA-731:K29Q
33	33	ligand_near	25.5	OXA-1307:A33S; OXA-54:A33T
34	34	ligand_near	22.3	OXA-923:H34Y; OXA-934:H34Y
36	36	surface_other	23.9	OXA-436:T36S; OXA-535:T36S
38	38	ligand_near	21.4	OXA-199:H38Y
40	40	surface_other	19.0	OXA-436:S40T; OXA-535:S40T
44	44	surface_other	17.0	OXA-54:V44I
45	45	surface_other	16.5	OXA-199:V45A; OXA-252:V45A
51	51	surface_other	27.3	OXA-436:K51T; OXA-535:K51T
55	55	surface_other	20.8	OXA-731:F55Y
58	58	surface_other	20.7	OXA-436:N58D; OXA-535:N58D
59	59	surface_other	21.5	OXA-1309:L59I
65	65	surface_other	12.9	OXA-1308:A65V
75	75	catalytic_near	6.8	OXA-918:P75T
98	98	surface_other	15.9	OXA-204:Q98H
99	99	surface_other	16.3	OXA-204:T99R
104	104	ligand_near	11.4	OXA-1119:T104A; OXA-1181:T104A
110	110	surface_other	12.8	OXA-1119:N110D; OXA-1181:N110D
120	120	catalytic_near	4.2	OXA-519:V120L; OXA-788:V120L
121	121	catalytic_near	8.6	OXA-1304:P121A
123	123	catalytic_near	7.1	OXA-439:Y123H
125	125	surface_other	12.4	OXA-245:E125Y; OXA-535:E125G
127	127	surface_other	11.9	OXA-1242:A127S
129	129	surface_other	16.9	OXA-731:Q129K
132	132	surface_other	17.3	OXA-54:E132Q; OXA-731:E132Q
135	135	surface_other	12.1	OXA-1305:M135I
136	136	surface_other	16.9	OXA-731:S136N
141	141	surface_other	18.0	OXA-566:A141D
153	153	surface_other	12.6	OXA-416:V153L; OXA-436:V153L
154	154	surface_other	10.7	OXA-199:D154G
155	155	surface_other	9.3	OXA-1038:S155T; OXA-416:S155T

Table 5: OXA-48-like variant positions, structural annotation.

pos	pdb res	zone	d-S70	examples
159	159	surface_other	9.2	OXA-933:D159N
167	167	surface_other	15.1	OXA-1038:T167I; OXA-1226:T167I
168	168	surface_other	13.3	OXA-1119:E168Q; OXA-1181:E168Q
170	170	surface_other	13.4	OXA-54:I170V
171	171	surface_other	15.4	OXA-1039:S171T; OXA-1119:S171A
201	201	surface_other	25.8	OXA-416:G201A; OXA-436:G201A
211	211	catalytic_near	4.6	OXA-247:Y211S
212	212	catalytic_near	6.1	OXA-1211:S212L; OXA-1240:S212G
213	213	ligand_near	8.2	OXA-1119:T213I; OXA-162:T213A
214	214	ligand_near	9.2	OXA-1167:R214S; OXA-1201:R214S
215	215	surface_other	11.4	OXA-438:I215Y
216	216	surface_other	13.8	OXA-438:E216D; OXA-517:E216K
217	217	surface_other	11.6	OXA-1012:P217S; OXA-1055:P217S
218	218	surface_other	11.6	OXA-1212:K218N; OXA-54:K218Q
219	219	ligand_near	8.2	OXA-538:I219F
226	226	surface_other	16.8	OXA-436:V226I
237	237	surface_other	10.7	OXA-436:M237T; OXA-535:M237T
244	244	ligand_near	12.5	OXA-1181:S244W; OXA-1205:S244W
245	245	ligand_near	15.2	OXA-436:D245E; OXA-535:D245E
252	252	ligand_near	16.3	OXA-436:A252T; OXA-535:A252S
256	256	ligand_near	20.8	OXA-436:E256A; OXA-535:E256A
259	259	surface_other	23.3	OXA-833:K259T

C Novel variant calls with amino-acid changes or stops (2023+)

92 of 357 characterized recent non-exact calls alter protein sequence or length; the remaining 265 are synonymous at protein level or partial.

Table 6: Protein-altering novel NDM/OXA-48-like calls.

assembly	closest ref	changes (n)	stops
GCA_031038655.3	NDM-3	E152K (1)	0
GCA_031056435.1	NDM-1	D130N M154L (2)	0
GCA_032671545.1	NDM-1	del257A del258A del259I del260T del261H del262T (15)	0
GCA_032868935.2	NDM-5	V155G (1)	0
GCA_037887805.1	NDM-5	G29D (1)	0
GCA_038419775.1	NDM-1	ins10S ins10E ins11E ins11H K12R L13I T15R A16C (226)	6
GCA_038487205.1	NDM-1	M154L (1)	0
GCA_040442885.1	NDM-4	S249G (1)	0
GCA_040442885.1	OXA-181	ins0L ins0N M1Y V3D L4E A5V A8C V9A ... (213)	12
GCA_042177995.1	NDM-1	V88L M154L (2)	0
GCA_044895485.1	NDM-1	V88L M154L (2)	0
GCA_045445595.1	NDM-1	E152G (1)	0
GCA_045445635.1	NDM-1	E152G (1)	0

Table 6: Protein-altering novel NDM/OXA-48-like calls.

assembly	closest ref	changes (n)	stops
GCA_045445675.1	NDM-1	E152G (1)	0
GCA_045445755.1	NDM-1	E152G (1)	0
GCA_045718445.1	NDM-5	del256R del257A del258A del259I del260T del261H (16)	0
GCA_046367555.1	NDM-5	del219G N220X L221X G222X D223X A224R D225T T226 (48)	0
GCA_046368305.1	NDM-5	P9S (1)	0
GCA_047893855.1	OXA-232	A104T D110N Q168E A171S S214R (5)	0
GCA_047893895.1	NDM-4	M1N E2S L3A (3)	0
GCA_047894095.1	NDM-4	M1N E2S L3A (3)	0
GCA_047894795.1	NDM-4	M1N E2S L3A (3)	0
GCA_047919235.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_048020515.1	NDM-4	M1N E2S L3A (3)	0
GCA_048247675.1	NDM-1	V88L M154L (2)	0
GCA_049222765.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_049232105.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_049236645.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_049569975.1	NDM-1	V88L M154L (2)	0
GCA_050379495.1	NDM-1	H8P P9V (2)	0
GCA_050582225.1	NDM-5	del262T del263A del264R del265M del266A del267D (10)	0
GCA_050595725.1	NDM-1	V88L M154L (2)	0
GCA_050595745.1	NDM-4	V88L (1)	0
GCA_050690955.1	NDM-5	del262T del263A del264R del265M del266A del267D (10)	0
GCA_050722475.1	NDM-5	M265I (1)	0
GCA_050724915.1	NDM-5	D225E (1)	0
GCA_051205915.1	NDM-5	T162I (1)	0
GCA_051414975.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_052445325.1	NDM-4	V88L (1)	0
GCA_052568655.1	NDM-1	E2D (1)	0
GCA_052594815.1	NDM-5	A243T (1)	0
GCA_052942835.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_052942855.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_052942875.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_052942895.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_052942915.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_052942935.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_052942955.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_052942975.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_052942995.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_052943015.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_052943055.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_053194025.1	NDM-5	G29D (1)	0
GCA_053336735.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_053336755.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_053337215.1	NDM-5	M1A E2P L3G P4R N5E I6A M7E del9P ... (231)	6
GCA_053366035.1	NDM-1	E2D (1)	0
GCA_053513495.1	NDM-5	del146N Q147G L148I A149G P150S Q151I E152N G153 (112)	8

Table 6: Protein-altering novel NDM/OXA-48-like calls.

assembly	closest ref	changes (n)	stops
GCA_053848185.1	NDM-5	G36S (1)	0
GCA_053848345.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_054177645.1	NDM-1	del73V A74C S75R N76F G77Q L78R I79F V80D ... (175)	5
GCA_054488385.1	NDM-1	I137V (1)	0
GCA_054792955.1	NDM-1	del1M E2* L3M P4G N5S I6C M7A H8A ... (231)	6
GCA_054792955.1	NDM-5	del1M E2* L3M P4G N5S I6C M7A H8A ... (229)	6
GCA_054794435.1	NDM-71	D110N (1)	0
GCA_055434245.1	NDM-5	M245I (1)	0
GCA_056740755.1	NDM-1	Q123K M154L (2)	0
GCA_056781405.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_056969255.1	NDM-4	D130N (1)	0
GCA_057174215.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_057412325.1	NDM-1	M1* E2P ins3H ins3* ins3C P4* N5A I6G ... (228)	10
GCA_057412325.1	NDM-5	M1* E2P ins3H ins3* ins3C P4* N5A I6G ... (228)	10
GCA_057731325.1	NDM-5	del1M E2H L3* del5N I6L M7H H8* P9C ... (230)	10
GCA_057961985.1	NDM-5	M1K E2C L3R N5L I6N M7G H8F P9M ... (28)	2
GCA_058143285.1	NDM-5	R264S (1)	0
GCA_058143815.1	NDM-1	V88L M154L (2)	0
GCA_058195915.1	NDM-5	D130G (1)	0
GCA_058313035.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_058666915.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_059020645.1	NDM-5	E108V (1)	0
GCA_059034205.1	NDM-5	D130G (1)	0
GCA_059621975.1	NDM-1	W104R A149* P150R Q151G E152D M154C V155G del157 (11)	1
GCA_059637815.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_059637835.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_059682985.1	NDM-1	Q123K M154L (2)	0
GCA_059932675.2	NDM-1	del211K D212W S213T K214A A215R K216P L218R G219 (52)	3
GCA_059990755.1	NDM-1	A263V (1)	0
GCA_061007145.1	NDM-1	Q158H (1)	0
GCA_061012805.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_061073645.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6
GCA_061369735.1	NDM-5	del245M del246I del247V del248M del249S del250H (27)	0
GCA_977071415.1	NDM-1	M1A E2P L3G P4R N5E I6A M7E del9P ... (232)	6

D Geographic and host-species distribution (2023+)

E OXA-48-like family definition

The 67 family members with amino-acid identity to OXA-48 (global alignment, full length).

Table 7: Top countries of isolation, recent isolates.

NDM country	**n**	**OXA-48-like country**	**n**
USA	26,557	USA	4,203
China	3,073	Slovenia	1,237
Netherlands	1,252	Netherlands	1,119
India	1,091	South Africa	692
Slovakia	758	Denmark	632
Canada	721	Spain	600
Italy	661	India	467
Slovenia	594	Slovakia	450
Germany	437	Thailand	323
France	328	Germany	288

Table 8: Top organisms, recent isolates.

NDM organism	**n**	**OXA-48-like organism**	**n**
Klebsiella pneumoniae	18,066	Klebsiella pneumoniae	7,352
E.coli and Shigella	11,688	E.coli and Shigella	4,367
Enterobacter hormaechei	3,319	Citrobacter freundii	501
Acinetobacter baumannii	2,167	Klebsiella oxytoca	239
Providencia alcalifaciens	1,365	Enterobacter hormaechei	122
Pseudomonas aeruginosa	980	Acinetobacter baumannii	65
Citrobacter freundii	764	Providencia alcalifaciens	54
Klebsiella oxytoca	569	Serratia marcescens	24

Table 9: OXA-48-like family membership.

allele	**identity**	**allele**	**identity**
OXA-48	1.000	OXA-162	0.996
OXA-244	0.996	OXA-245	0.996
OXA-252	0.996	OXA-370	0.996
OXA-505	0.996	OXA-519	0.996
OXA-566	0.996	OXA-894	0.996
OXA-918	0.996	OXA-920	0.996
OXA-933	0.996	OXA-934	0.996
OXA-1039	0.996	OXA-1167	0.996
OXA-1212	0.996	OXA-1242	0.996
OXA-1306	0.996	OXA-204	0.993
OXA-514	0.993	OXA-515	0.993
OXA-546	0.993	OXA-1038	0.993
OXA-1213	0.993	OXA-1304	0.993
OXA-793	0.990	OXA-1146	0.990
OXA-199	0.989	OXA-416	0.989
OXA-538	0.989	OXA-547	0.989
OXA-517	0.986	OXA-1055	0.986
OXA-181	0.985	OXA-567	0.983
OXA-1012	0.982	OXA-405	0.982
OXA-1200	0.982	OXA-232	0.981
OXA-484	0.981	OXA-923	0.981
OXA-924	0.981	OXA-1119	0.981

Table 9: OXA-48-like family membership.

allele	identity	allele	identity
OXA-1181	0.981	OXA-1211	0.981
OXA-1226	0.981	OXA-1305	0.981
OXA-1308	0.981	OXA-163	0.978
OXA-929	0.978	OXA-1240	0.978
OXA-833	0.977	OXA-922	0.977
OXA-1201	0.977	OXA-1205	0.977
OXA-1207	0.977	OXA-1307	0.977
OXA-1309	0.977	OXA-438	0.976
OXA-247	0.974	OXA-439	0.974
OXA-788	0.974	OXA-54	0.924
OXA-436	0.913	OXA-535	0.913
OXA-731	0.913		

F Co-carriage combinations

5,803 assemblies carry both blaNDM and blaOXA-48-like genes. Top combinations:

Table 10: NDM x OXA-48-like co-carriage (assemblies).

NDM allele	OXA-48-like allele	assemblies
blaNDM-5	blaOXA-181	1,915
blaNDM-1	blaOXA-48	1,747
blaNDM-1	blaOXA-232	1,482
blaNDM-5	blaOXA-48	1,456
blaNDM-5	blaOXA-232	1,143
blaNDM-1	blaOXA-181	385
blaNDM-4	blaOXA-181	332
blaNDM-5	blaOXA-1207	119
blaNDM-5	blaOXA-244	108
blaNDM	blaOXA-48	98
blaNDM-5	blaOXA-1205	95
blaNDM-5	blaOXA-484	80
blaNDM	blaOXA-181	72
blaNDM	blaOXA-232	66
blaNDM-4	blaOXA-48	48

G Data provenance and query record

All population data were pulled 2026-09-23 from the NCBI Pathogen Detection isolate backend (the solr API behind the public Isolates Browser), collection “amr” (AMRFinderPlus calls; refgene version recorded per row) and collection “isolates” (SNP-cluster join, 100-accession batches). Reference data: CARD latest data tarball (4,366,637 bytes, sha256 94b89501fc26daafbe65e4151c17a8b007dc1272c4f8201737e4
NCBI beta-lactamase allele registry Allele.tab/Allele-dna.fa (1,557 alleles). Structures: RCSB PDB 3SPU, 4EYL, 4S2P, 6P97 downloaded 2026-09-23. Assemblies: NCBI Datasets v2 API,

GENOME_FASTA, 711 of 722 targets delivered. Software: blastn 2.17.0+, Python 3.10.12, Biopython 1.88. Manifest: ndm-oxa/data/MANIFEST.sha256 (44 payloads).

H Gate evidence

G1: 136/136 reference self-recovery (blastn, CARD reference DB). G2: 1,599/1,607 (99.50%) vs NCBI AMRFinderPlus; 8 disagreements resolved. G4: spike-in controls detected; published-genome control (run DRR065574) passed with both documented disruptions recovered at exact positions and a clean negative control (data/g4/g4_evidence.json; data/g4/G4_MANIFEST_ADDENDUM.sha256).

I Verified/thin/missing summary

Verified: allele calls (G1/G2-backed), stop-codon porin disruptions, structural distances from PDB coordinates, H250N and L10-loop novel variants (sequence-verified). Thin: porin indel classes (loop-polymorphism confound), OXA-48-like borderline family members, GCA_059932675.2 L10-loop cluster (possible sequencing artifact). Missing: 11 undelivered assemblies, MIC phenotypes (not in Pathogen Detection).